

Payoff scale reshapes how language models play the prisoner’s dilemma

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Abstract

Multiplying every payoff in a game by a positive constant leaves preferences, best replies, equilibria and the orbits of the replicator dynamics untouched, so the predicted effect of that rescaling is zero by theorem rather than by convention. Whether language models respect that invariance is untested. To use the invariance as an instrument, we had five frontier language models play an iterated prisoner’s dilemma across ten payoff scales, crossed with prompt language and persona instruction, on a complete, balanced grid of 200,000 decisions. Cooperation moves in every model, by up to 0.243 (95% CI 0.200, 0.290). The movement has a different shape in each, so averaging over models reports it as almost absent, and its size depends on the prompt language. The scale also gates the persona instruction, reversing its sign in two models of five where every printed payoff is at most 1. In three models of five what is played changes too, becoming less consistent with any canonical memory-one rule, while a fourth moves the other way. The effect is already present on the opening move. These departures suggest that cooperation rates reported for language models should be interpreted together with their sensitivity to payoff scale.

Keywords: machine behaviour, large language models, iterated prisoner’s dilemma, payoff scaling, utility invariance, benchmark validity

1 Introduction

Language models are increasingly deployed as agents that negotiate, coordinate and decide on behalf of their users [1, 2], and a fast-growing literature measures how they behave when they do: through behavioural economics and cognitive psychology [3–5], and through the classical games, where a chatbot’s play can be hard to tell from a human’s [6]. The prisoner’s dilemma is its workhorse, because its strategy space is mapped and its predictions are sharp [7–12]. Within it the models turn out to be individual rather than uniform. GPT-4 defects for good after a single defection yet fails to coordinate where the game rewards alternation [13]; other models forgive at rates no human population shows [14]; the weight a model gives the structure of the game rather than the story around it differs within a single generation [15]; and a model told that it is selfish moves in the instructed direction without carrying it into conditioned reciprocity [16]. These studies share a reporting convention that goes unremarked. Each summarizes a model by a cooperation rate, or a small set of them, measured at one payoff matrix.

Against that convention sits a body of work showing that what is measured on a language model depends on things the measurement was not about. Accuracy on a fixed task moves with the order of the in-context examples [17], and with formatting choices that leave the meaning of the prompt intact, by margins wide enough to swamp the differences between the models being compared [18]. Instructions about who the model is are equally consequential: a persona changes what a model produces and what it will produce at all [19], shifts measured capability [20], and, conditioned on a demographic backstory, reproduces the response distribution of the matching human subgroup [21]. What none of these manipulations can supply is a null. Reordering examples is genuinely different evidence, a new persona is genuinely new information, and a different language is a different training distribution.

The choice of units is not usually stated as a design decision at all, because in the theory it is not one. A payoff matrix is a von Neumann–Morgenstern utility, defined only up to a positive affine transformation [22, 23], so writing the same dilemma with cells ten times larger changes no preference, no best reply, no equilibrium and no orbit of the replicator dynamics. Under the theory the two matrices are not two games written differently. They are one game. The theory of the iterated dilemma inherits the indifference: its tournament results, memory-one rules and zero-determinant strategies turn on the ordering of the four payoffs and on ratios among them, never their absolute size [7, 9, 10, 12], and the modelling literature normalizes the scale away as a matter of course [24, 25]. Current theory works at the indifference rather than merely permitting it: where social learners update on the payoff they last received instead of on an expected payoff, the condition for conditional cooperation to persist under strong selection loses the benefit and the cost of cooperation altogether and turns on the ordering of the payoffs alone [26].

The invariance belongs to the theory, not to people: human valuation is neither linear in magnitude nor anchored at zero [27], and human choice reverses between logically equivalent descriptions of one problem [28]. That gap has since been argued to be structural rather than incidental, with the linguistic description of an action entering the utility function directly and a shift proposed from outcome-based to language-based preferences [29]. Which is why it is worth putting the question to a machine asked to play the game as written.

Two things are established, and they have not been put together. The first is that what a language model does depends on features of the prompt that carry no task content: the order of the examples, the formatting, the persona, the language. The second is that multiplying every payoff by a positive constant carries no task content in the strictest sense available, because the matrix it produces is not a second game but the same one. What is not known is whether the strategic behaviour of a language model respects that invariance. To our knowledge, no study has asked. Every manipulation in the sensitivity literature could legitimately change behaviour, so a measured change under any of them is at worst a nuisance and at best a finding, against an expected effect nobody had specified. The same holds of the human framing results that motivate language-based preferences, where a label such as donate rather than boost carries connotation, so what is under test there is an assumption about preferences and not a theorem about their representation [29]. A rescaling is the one manipulation whose expected effect is specified, and specified as zero, so a measured departure has a stated reference value rather than a comparison condition.

What follows if the invariance fails is a question about what a published cooperation rate means. Machine behaviour proposes that autonomous systems be studied empirically, with

the instruments of the behavioural sciences, because what such a system does is not deducible from how it was built [30]. Those instruments arrive with a known difficulty: a behavioural manipulation almost never carries a predicted effect, so a measured change has to be read against a comparison condition the experimenter chose. The invariance classes of decision theory are the exception, supplying a manipulation whose predicted effect is a stated number, and the number is zero.

The consequences run through the strands of machine behaviour this study touches. The persona line is the one place where the experimenter addresses the agent about its social role, and whether it takes hold turns out to depend on the units the matrix was printed in. The departure is already present on the opening move, which locates it in the reading of the matrix rather than in the weighing of a history. We report no evolutionary dynamics of our own; what we report is a constraint on an input such dynamics take, since models of mixed human-machine populations carry the cooperativeness of the artificial agent as a parameter and ask where the population settles [31–35]. A rate partly determined by the experimenter’s units is not the quantity those models need, and if the dependence takes a different form in each model it is invisible to exactly the pooled designs the field uses to summarize many models at once.

We take the rescaling as our object of study, and ask four questions of it. Does strategic behaviour survive multiplying every payoff by a positive constant, and by how much does it fail to? Does the answer depend on the prompt language and on the persona the model is told to adopt? Does the rescaling change only how often a model cooperates, or the strategy it is playing? And does the effect enter at the opening move, before any history exists, or through the course of play? We answer them on a complete and exactly balanced grid of 10,000 dyads, set out in §2.

Here, we turn a theorem into an instrument, and report three things. First, a positive rescaling of the payoffs has a predicted behavioural effect of exactly zero, and we use it to ask whether five frontier language models are playing the game they were shown: all five depart from the invariance, by up to 0.243 in cooperation. Second, we show that the departure has a different shape in each model, to the point of anticorrelation, so that pooling models returns no significant main effect of the payoff scale while every model in the pool is moving substantially; the same holds one level down, since the size of the departure depends on the prompt language as well as on the model. Per-model, per-language reporting is therefore a matter of validity rather than of presentation. Third, we show that the departure is structural and not a shift in level: the payoff scale gates the persona instruction, reversing its sign in two models of five and abolishing it in a third, play becomes measurably less consistent with any canonical memory-one rule in three models of five while a fourth moves significantly the other way, and the effect is already present on the opening move, before any history exists.

2 Material and methods

2.1 Experimental design

The design is a complete factorial crossing of five language models, ten payoff scales, five prompt languages, four persona pairings and ten replicates, at ten rounds per game and two agents per game.

The unit of collection is the dyad, one ten-round game between two agents of the same

model. It contributes two agent-games, an agent-game being the ten decisions one agent takes in one game, and twenty decisions in all. Each of the 250 model-by-language-by-scale cells holds exactly 40 dyads, the four persona pairings by ten replicates, hence 80 agent-games; pooling the five languages gives 200 dyads and 400 agent-games in each of the 50 model-by-scale cells. The corpus is 10,000 dyads, 20,000 agent-games and 200,000 decisions.

Balance is enforced rather than assumed. The build refuses to write its tables if a cell departs from 40 dyads, if an agent-game is not ten rounds long, or if the payoff scale recovered from the payoffs recorded in a game disagrees with the scale the cell was run at (electronic supplementary material, §S1).

2.2 The game and the payoff transformation

Both agents receive the prisoner’s-dilemma prompt of the FAIRGAME protocol [36] in its no-communication, announced-horizon condition. It casts the game as a criminal interrogation, states the four payoffs as years of imprisonment, names the two actions OptionA and OptionB, carries the running history of play, and instructs the agent to minimize its penalty.

At $\lambda = 1$ the agent is shown $T = 0$, $R = 2$, $P = 6$, $S = 10$ in penalty units, with T the temptation, R the reward for mutual cooperation, P the punishment for mutual defection and S the sucker’s payoff. The ordering $T < R < P < S$ is the image of the familiar $T > R > P > S$ on utilities $u = -\text{penalty}$, and $2R > T + S$ holds on those utilities, so alternation is not preferred to mutual cooperation. OptionA pays less in both columns and is therefore the dominant, defecting action, while mutual OptionB at 2 and 2 beats mutual OptionA at 6 and 6 and is the cooperative one. $T = 0$ is the penalty for taking OptionA against OptionB, and $S = 10$ that of the player who took OptionB alone. Cooperation means OptionB throughout this paper. The polarity was checked against data as well as against the prompt: reproducing a published cooperation curve from these tables gives a mean absolute error of 0.002 under this mapping and 0.171 under the inverted one [37].

The manipulation multiplies all four cells by a positive constant λ (figure 1a), taking ten values spanning five orders of magnitude, $\lambda \in \{0.01, 0.1, 0.25, 0.5, 1, 2, 5, 10, 100, 1000\}$, and changes nothing else in the prompt. Products print as integers wherever they are integral, so at $\lambda = 1$ the prompt is character-for-character the conventional one. Two details bear on interpretation: the cell $T = 0$ prints as 0 at every scale, so only three of the four numbers move; and the history block records realised penalties, so λ is present in every prompt of the game rather than only the first. We treat the printed penalties as the agent’s utilities, which is what the prompt’s minimization instruction directs it to do. On that premise, and because λu with $\lambda > 0$ is a positive affine transformation of a von Neumann–Morgenstern utility [22, 23], the predicted effect on an agent playing the game is exactly zero. An agent that instead applies a non-linear valuation to years of imprisonment is not bound by the invariance; §4 returns to readings of that kind.

2.3 Models, prompt languages and personas

Five frontier models were run, each against itself; two different models were never paired. The models, with the endpoint identifier of each in parentheses, were Claude Haiku 4.5 (claude-haiku-4-5-20251001), GPT-5.4 Nano (gpt-5.4-nano-2026-03-17), Gemini 3.1 Flash-Lite (gemini-3.1-flash-lite-preview), Gemini 3.5 Flash-Lite (gemini-3.

5-flash-lite) and Qwen3 235B-A22B (qwen3-235b-a22b-instruct-2507); the short names are the ones used in the figures and tables throughout. All five were served through the Kaggle Benchmarks model proxy on an OpenAI-compatible chat-completions endpoint (electronic supplementary material, §S2). The two Gemini identifiers name endpoints that are not pinned to a release date.

The prompt languages are English, French, Vietnamese, Chinese and Arabic. The templates are byte-identical copies of the FAIRGAME resource files and are compared byte for byte against them before any collection run (electronic supplementary material, §S2). Two artefacts of those files are reproduced rather than repaired: the Arabic and Chinese versions state the four cells as penalties but phrase the goal sentence as maximizing a reward, and the Vietnamese version doubles an article in the persona line. We therefore treat the five languages as five distinct stimuli rather than five renderings of one.

Each agent also receives one line naming its persona, in the language of the prompt: in English, *You are cooperative.* or *You are selfish.* The two instructions are crossed in all four ordered pairings, ten replicates each. The opponent’s persona is never disclosed (electronic supplementary material, §S2), so a pairing reaches a player only through the opponent’s realised behaviour, and not before round 2, which is why we treat it as a design factor and a regression covariate rather than as information the agent held.

2.4 Protocol and data collection

Every game runs ten rounds and the horizon is announced, in English as *There are 10 rounds to decide.* Moves are simultaneous: both agents see the same history, rounds 1 to $r - 1$, and the round is scored once both have replied.

2.4.1 Common random numbers

The seed sent with each request is a deterministic function of a fixed base seed (12345) and an index that excludes λ by design, so the ten scales share one stream of seeds and a given round of a given replicate carries the same seed at every scale. Sampling runs at temperature 1.0 and a hosted endpoint honours a seed on a best-effort basis, so the scheme is an attempt at variance reduction rather than determinism, and its achieved effect is not measured here (electronic supplementary material, §S2.1).

2.4.2 Elicitation and parsing

Decisions are elicited as free text, with no response schema and no function calling, under a ceiling of 128 output tokens and no reasoning-effort parameter. Replies are parsed by the protocol’s own matcher (electronic supplementary material, §S2.2).

2.4.3 Unparseable replies

A reply the matcher cannot resolve is regenerated with a different seed, up to two further attempts. If all three fail, the decision is recorded as OptionA, which under this framing is defection: the fallback can only push a measured cooperation rate down, never up, and it cannot manufacture the cooperative behaviour reported here. Each collection run asserts that its fallback rate is at most 0.02 and fails otherwise. The achieved rates are held in the collection logs

rather than in the released per-game tables, so we cannot report how they vary with the payoff scale; a rate that rose with the printed magnitudes would bias the smallest ranges downward.

2.5 Statistical analysis

2.5.1 Cooperation rate and uncertainty

The cooperation rate of an agent-game is the share of its ten decisions that are OptionB; a cell mean averages the agent-games of a cell, and round-level analyses use the binary decision itself. The two agents of a dyad play each other and the ten rounds inside an agent-game are strongly autocorrelated, so intervals computed as though the 200,000 decisions were independent would be far too narrow. Every interval was computed by resampling whole dyads with replacement, 1,500 draws, percentile method at the 2.5th and 97.5th centiles [38], and every regression clustered its standard errors on the dyad. One seed drove every resampling in the paper (electronic supplementary material, §S2.3). Because the range over the ten scales is a maximum minus a minimum, and so positive by construction, its bootstrap interval quantifies the precision of the range and not its distance from zero; the permutation test below supplies the test.

2.5.2 Permutation test

Because the null is a theorem, the matching test destroys the association with the payoff scale and leaves the rest of the design as observed. We shuffled the scale label among whole dyads within strata of prompt language by persona pairing inside each model, and recomputed the range of the ten scale means. The dyad is the unit shuffled because the payoff scale was assigned to a dyad, so under the null both of its agent-games carry the same label; shuffling agent-games independently would break that pairing and draw from a null narrower than the true one (electronic supplementary material, §S2.3). With 2,000 draws and a p -value of $(\text{hits} + 1)/(\text{draws} + 1)$ the smallest attainable value is 0.0005.

2.5.3 Per-model regression

For each model we fitted a logistic regression of the round-level decision on the payoff scale as a ten-level factor, with prompt language, own persona and opponent persona as covariates, $\lambda = 1$ as the reference level and standard errors clustered on the dyad: $n = 40,000$ decisions in 2,000 clusters per model. The reported test is a Wald test on the nine scale contrasts jointly.

2.5.4 Variance decomposition

A type-II analysis of variance on the 250 cell means, with terms for model, language and payoff scale and their three two-way interactions. Cell means rather than agent-games, because the design is exactly balanced at the cell level, which makes the type-II decomposition order-independent. There is no three-way term, so the residual on 144 degrees of freedom is the omitted three-way interaction and every F is tested against it.

2.5.5 Regimes and trends

With base payoffs 0, 2, 6, 10 the largest number the prompt prints is 10λ , so every payoff is at most 1 exactly when $\lambda \leq 0.1$, which two of the ten scales satisfy. The persona result contrasts

those two scales with the remaining eight, as a difference of differences with an interval from the same dyad bootstrap. Where a trend across the whole grid is wanted instead, $\log_{10} \lambda$ enters as a continuous predictor, because the transformation is multiplicative.

2.5.6 Software

Analyses were run in Python 3.12.3 with `numpy` 1.26.4, `pandas` 2.2.2 and `statsmodels` 0.14.6 for the clustered logistic regressions and the type-II analysis of variance, `matplotlib` 3.9.2 for the figures, and `pytorch` 2.11.0 for the classifier described below.

2.6 Strategy read-out

The read-out reports which of four canonical rules a ten-round trajectory most *resembles*. It is not an identification of the strategy a model is running, and no claim in this paper treats it as one.

2.6.1 Exact matching

The four canonical rules, AllC, AllD, Tit-for-Tat and Win-Stay-Lose-Shift, are all memory-one, so asking whether a trajectory is precisely what a rule would have played is a per-token lookup: no learning, and no counterfactual simulation. The answer is a *set* rather than a label, because over ten rounds several rules routinely prescribe the same moves and no method can separate them on that evidence. Three provenances, meaning the three ways in which a label can have been arrived at, are therefore recorded separately and never merged: the agent-games matched exactly by one rule (deduced), those matched by several at once (ambiguous) and those matched by none (unmatched). The same matcher yields a continuous quantity needing no labels at all, the number of rounds violating the nearest rule.

2.6.2 The learned branch and arbitration

A single-layer LSTM [39] is trained on the synthetic corpus alone, which holds canonical trajectories of those four rules at four levels of execution noise and no language-model output whatever; its encoding, architecture and optimiser recipe are in the electronic supplementary material, §S6. Training uses the noise-free and 5% levels, and the 10% and 20% levels are held out entirely as a check on how the read-out degrades as play departs further from any rule. Accuracy is 98.5% on the held-out split, 93.6% at the unseen 10% level and 81.1% at 20%. The rule base decides first and the classifier ranks only within whatever set it returns, so it can never overturn a deduction; it decides alone only where no rule fits. Its share of the labels is very unevenly spread across the four rules, which bounds how the composition may be read (§3).

3 Results

Across the 200,000 recorded decisions cooperation is 0.565.

3.1 Cooperation moves under a transformation that cannot matter

We find that cooperation depends on the payoff scale in all five models (figure 1b, table 1). The range across the ten scales runs from 0.072 (95% CI 0.060, 0.133) in Gemini 3.1 Flash-Lite to

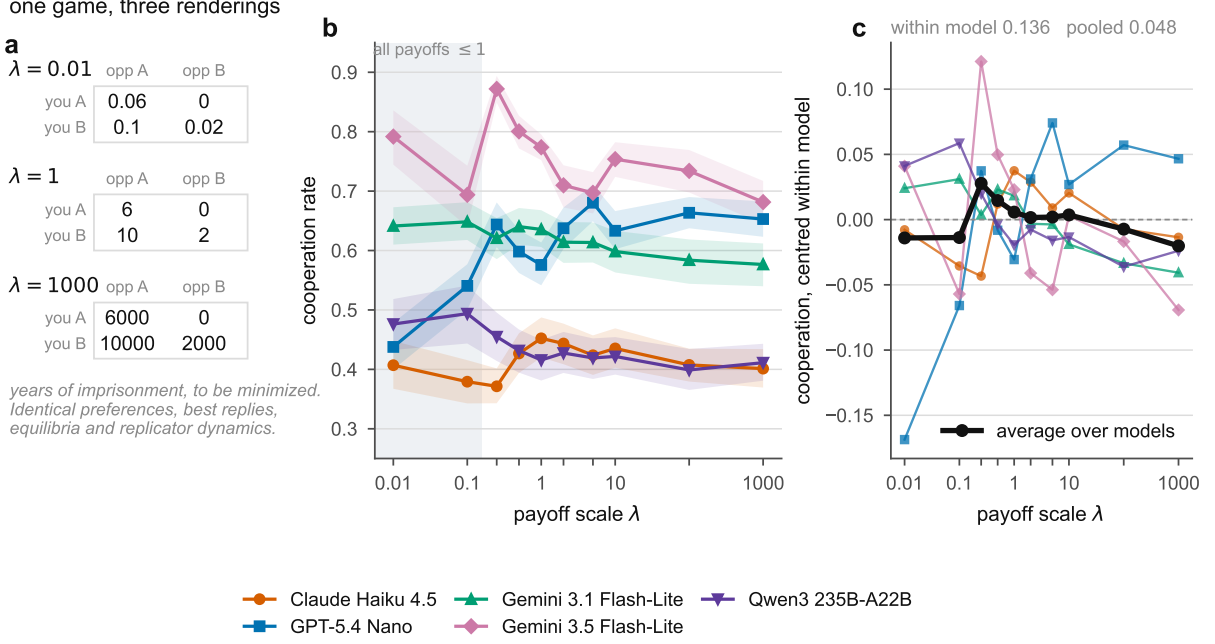


Figure 1: Cooperation depends on the payoff scale in all five models. (a) The same prisoner’s dilemma at three payoff scales, shown to the agent as years of imprisonment to be minimized. Multiplying every cell by λ leaves preferences, best replies, equilibria and the orbits of the replicator dynamics unchanged. (b) Cooperation against the payoff scale for each model, with 95% confidence intervals from a bootstrap over whole dyads. The shaded band marks the two scales at which every payoff printed is at most 1. (c) The same curves centred within model, with their average in black. The average varies by 0.048 across the ten scales, where the mean within-model variation is 0.136. Each point in (b) and (c) averages 200 dyads; the intervals are percentile intervals from 1,500 bootstrap resamples.

0.243 (95% CI 0.200, 0.290) in GPT-5.4 Nano, with Gemini 3.5 Flash-Lite at 0.190 (95% CI 0.165, 0.240); each interval is on the range rather than on the cooperation rate, and resamples whole dyads (§2).

In the permutation test of §2 the observed range falls in the extreme upper tail of the permuted distribution in every model (table 1: $p = 0.0005$ in four of the five, the smallest value 2,000 draws can return, and $p = 0.0025$ in Claude Haiku 4.5). The per-model logistic regression rejects the joint null on the nine scale contrasts in every model (table 1; $\chi^2_9 = 32.3$ to 164.1, all $p \leq 2 \times 10^{-4}$), and the largest contrast against $\lambda = 1$ ranges from 0.36 to 0.74 in log-odds.

The shapes are unlike one another. GPT-5.4 Nano rises steeply from 0.438 at $\lambda = 0.01$ and saturates near 0.65; Gemini 3.1 Flash-Lite drifts downward across the whole range; Claude Haiku 4.5 dips to a minimum at $\lambda = 0.25$ and recovers; Gemini 3.5 Flash-Lite oscillates within a high band.

That disagreement has a direct consequence for how such measurements are reported. Averaging the five model curves gives a pooled curve whose range is 0.048, while the mean within-model range is 0.136, an attenuation of 2.8 times (figure 1c). The pairwise correlations between model curves average -0.10 and run from -0.72 to $+0.73$, with six of the ten pairs negative. Curves that move in opposite directions cancel when added.

A variance decomposition on the 250 cell means makes the same point in the language a benchmark report would use. The model accounts for 56.5% of the variance across cells and the model-by-language interaction for 19.7%. The payoff scale as a main effect accounts for 0.7%

Table 1: Cooperation rate over the ten payoff scales, by model. Minimum and maximum cooperation rate over the ten scales, their range, a 95% confidence interval on that range from a bootstrap that resamples whole dyads, and a permutation test that shuffles the scale label within language and persona (2,000 draws, so 0.0005 is the smallest attainable value). The confidence interval is on the range. Ranges are computed before rounding, so they may differ by 0.001 from the difference of the rounded minimum and maximum. Each entry pools the five languages and the four persona pairings, 200 dyads per scale. The last two columns give a Wald test on the nine scale contrasts of a per-model logistic regression of the round-level decision, with standard errors clustered on the dyad ($n = 40,000$ decisions and 2,000 clusters per model).

Model	cooperation			95% CI	p_{perm}	Wald test	
	min	max	range			χ^2_9	p
Claude Haiku 4.5	0.371	0.452	0.081	[0.062, 0.135]	0.0025	32.3	< 0.001
GPT-5.4 Nano	0.438	0.680	0.243	[0.200, 0.290]	0.0005	164.1	< 0.001
Gemini 3.1 Flash-Lite	0.577	0.648	0.072	[0.060, 0.133]	0.0005	83.1	< 0.001
Gemini 3.5 Flash-Lite	0.682	0.872	0.190	[0.165, 0.240]	0.0005	129.0	< 0.001
Qwen3 235B-A22B	0.399	0.494	0.095	[0.068, 0.158]	0.0005	96.9	< 0.001

Table 2: Persona effect below and above the sub-unit boundary, by model. The persona effect is the cooperation rate of an agent told it is cooperative minus that of an agent told it is selfish. It is reported separately for the two scales at which every payoff printed is at most 1 ($\lambda \leq 0.1$) and for the eight at which some payoff exceeds it. Every interval is a 95% confidence interval from the dyad bootstrap. A sign flip is recorded only where both regime intervals exclude zero and fall on opposite sides of it, so a model whose supra-unit interval straddles zero is one whose persona effect is abolished rather than reversed.

Model	persona effect, with 95% CI		shift	95% CI	sign flip
	$\lambda \leq 0.1$	$\lambda \geq 0.25$			
Claude Haiku 4.5	−0.222 [−0.258, −0.186]	+0.207 [+0.189, +0.226]	0.429	[0.389, 0.471]	yes
GPT-5.4 Nano	−0.135 [−0.179, −0.091]	+0.018 [−0.001, +0.037]	0.153	[0.105, 0.200]	no
Gemini 3.1 Flash-Lite	+0.577 [+0.552, +0.602]	+0.680 [+0.667, +0.692]	0.103	[0.075, 0.131]	no
Gemini 3.5 Flash-Lite	−0.456 [−0.507, −0.406]	+0.224 [+0.207, +0.242]	0.680	[0.625, 0.736]	yes
Qwen3 235B-A22B	−0.911 [−0.931, −0.893]	−0.679 [−0.694, −0.664]	0.233	[0.210, 0.258]	no

and is not significant ($F_{9,144} = 1.10$, $p = 0.36$). Its interactions are: model by scale carries 6.5% ($p < 10^{-3}$) and language by scale 3.8% ($p = 0.041$), the last of which is the weakest test reported here and is not the evidence the language claim rests on. A study fitting a main effect for the payoff scale, pooling models and languages, will therefore be right about the average and wrong about every model in the pool.

3.2 The size of the scale effect depends on the prompt language

Neither of the two prompt factors that sit outside the game is neutral with respect to the payoff scale. We consider the prompt language first. Both the level of cooperation and its sensitivity to the scale differ with it (figure 2). Level differences are large and model-specific, which is why the model-by-language interaction dominates the decomposition above: Claude Haiku 4.5 cooperates 0.574 in Chinese and 0.234 in Vietnamese, while GPT-5.4 Nano cooperates 0.798 in English and 0.458 in Arabic.

Table 3: Rule agreement and rule distance against the payoff scale, by model. Both β columns are per decade of λ , and their sign is not the same in every model. The left block gives the share of agent-games matched exactly by one of the four canonical rules, below and above the sub-unit boundary, and the coefficient of a logistic regression of that indicator on $\log_{10} \lambda$. The right block gives the mean number of rounds that violate the nearest canonical rule at the smallest and largest scale, and the same trend fitted by least squares. Both use standard errors clustered on the dyad. Neither quantity depends on the learned classifier.

Model	exactly one rule fits (%)				rounds violating the nearest rule			
	$\lambda \leq 0.1$	$\lambda \geq 0.25$	β	p	$\lambda=0.01$	$\lambda=10^3$	β	p
Claude Haiku 4.5	17.9	3.5	-0.545	< 0.001	1.70	2.02	+0.073	< 0.001
GPT-5.4 Nano	9.8	7.6	-0.038	0.441	2.02	2.19	+0.038	0.052
Gemini 3.1 Flash-Lite	17.8	26.3	+0.125	< 0.001	1.08	1.03	-0.019	0.262
Gemini 3.5 Flash-Lite	32.2	15.6	-0.242	< 0.001	0.48	1.36	+0.194	< 0.001
Qwen3 235B-A22B	72.1	53.6	-0.267	< 0.001	0.27	1.14	+0.202	< 0.001

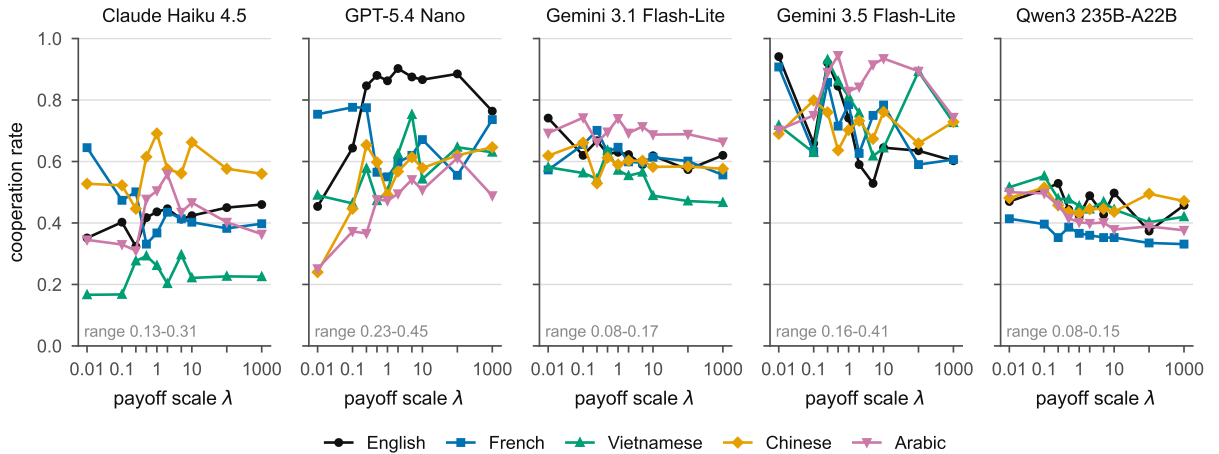


Figure 2: Cooperation against the payoff scale, by model and prompt language. One panel per model, one curve per prompt language. The annotation in each panel gives the smallest and largest across-scale range among that model’s five languages. Each point averages 40 dyads, pooling the four persona pairings and the ten replicates.

One caveat attaches to the level comparison rather than to the scale result. The Arabic and Chinese templates, reproduced unaltered from the protocol, state the four cells as penalties while phrasing the goal sentence as maximizing a reward (§2), so a level difference between two languages here is a difference between two stimuli and not between two renderings of one. It cannot produce a dependence on the payoff scale, because the wording is identical at every scale, and the ranges below are computed within a language.

The sensitivity differences are the new observation. The range of cooperation across the ten scales itself depends on the language: 0.136 in English against 0.314 in French for Claude Haiku 4.5, and 0.162 in Chinese against 0.412 in English for Gemini 3.5 Flash-Lite. Across all model-language pairs it runs from 0.080 to 0.449. Scale sensitivity is therefore a property of the model and the language together, not of the model alone.

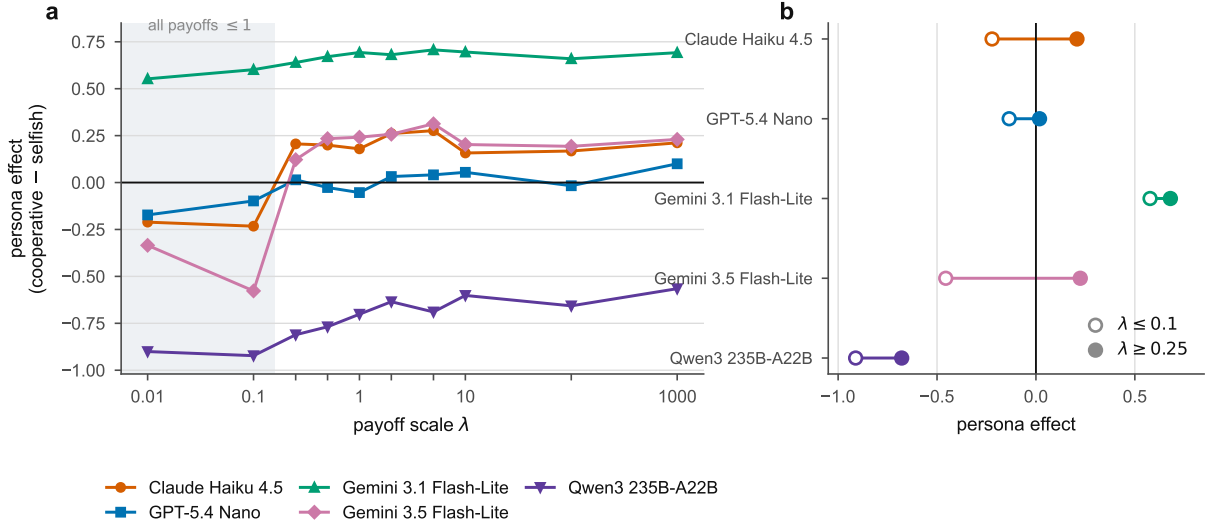


Figure 3: The persona effect changes with the payoff scale. (a) Persona effect, the cooperation rate under a cooperative persona minus that under a selfish one, against the payoff scale; each point averages 200 dyads. The shaded band marks the two scales at which every payoff printed is at most 1; the effect changes sign across it in two models of five (table 2). (b) The same contrast summarized: open markers give the mean persona effect at $\lambda \leq 0.1$, filled markers at $\lambda \geq 0.25$ (table 2).

3.3 The payoff scale gates the persona instruction

The difference between the cooperation rates of the two persona instructions measures how much the persona takes hold. It depends on the payoff scale, and not gradually (figure 3, table 2).

The natural boundary is where the printed numbers stop being sub-unit, at $\lambda \leq 0.1$ (§2). The persona effect shifts across it in all five models and every shift excludes zero, from 0.103 (95% CI 0.075, 0.131) in Gemini 3.1 Flash-Lite to 0.680 (95% CI 0.625, 0.736) in Gemini 3.5 Flash-Lite (table 2).

Whether the sign reverses is the stronger claim, and we hold it to a test on each regime rather than to a comparison of two point estimates. Two models of five reverse: in Gemini 3.5 Flash-Lite and Claude Haiku 4.5 the sub-unit interval lies wholly below zero and the supra-unit interval wholly above it, so below the boundary these models cooperate *more* when told they are selfish. In GPT-5.4 Nano the supra-unit interval straddles zero, so its negative sub-unit effect is abolished at the boundary rather than reversed. The other two models keep their sign throughout.

The boundary appears to be one of magnitude rather than of decimal notation. At $\lambda = 0.25$ the payoffs are 0, 0.5, 1.5, 2.5: still decimals, but no longer all below 1, and there the persona effect already has its supra-unit sign and size in both reversing models. The grid does not cross notation with magnitude fully, since every sub-unit scale in it is decimal-printed too.

Qwen3 235B-A22B inverts the persona instruction at every scale rather than only below the boundary (table 2). It appears to resolve the story’s ambiguity about who one cooperates with, one’s partner or the authorities, opposite to the convention the payoff matrix encodes. The inversion is not our subject, but its magnitude depends on the payoff scale like everything else measured here (electronic supplementary material, §S5).

3.4 The scale also changes what is played

Two models with the same cooperation rate can be playing different games, so we read each agent-game against the four canonical memory-one strategies (§2). What the read-out returns is a resemblance and not an identification, and the labels below should be read that way throughout. Over the whole corpus, 23.0% of agent-games are matched exactly by one rule, 13.2% by several at once and 63.7% by none, and the three provenances are reported separately throughout.

Play becomes less rule-governed as the payoffs grow in three models of five, and more so in a fourth (figure 4a, table 3). Pooling models, the share matched exactly by one canonical rule falls from 31.4% at $\lambda = 0.01$ to 19.5% at $\lambda = 10^3$, and the share matched by none rises from 54.4% to 69.2%. Per model the trend is significant and negative in Claude Haiku 4.5 ($\beta = -0.545$ per decade of λ), Qwen3 235B-A22B (-0.267) and Gemini 3.5 Flash-Lite (-0.242). It is significant and positive in Gemini 3.1 Flash-Lite ($+0.125$), and absent in GPT-5.4 Nano ($p = 0.44$).

The continuous version of the same measurement needs no labels at all: the number of rounds that violate the nearest canonical rule (figure 4b). It rises from 0.48 to 1.36 in Gemini 3.5 Flash-Lite and from 0.27 to 1.14 in Qwen3 235B-A22B between the smallest and largest scale, with significant positive trends there and in Claude Haiku 4.5. Being computed by exact rule matching, it does not depend on the classifier at all.

The composition shifts too, and the models do not shift together (figure 5). Pooled across models the movement is away from unconditional and towards reciprocal play, AllC falling from 45.6% to 40.4% of agent-games and Tit-for-Tat rising from 9.2% to 13.0% between the extreme scales, but that pooled figure conceals the same cancellation as before: GPT-5.4 Nano moves from 29.8% to 50.7% AllC while its AllD share falls from 46.0% to 17.5%, whereas Gemini 3.5 Flash-Lite moves the other way on AllC, from 73.8% to 51.5%, with Tit-for-Tat roughly doubling from 6.8% to 14.2%.

This last result leans hardest on the learned component, and the provenance is not uniform across the four labels: AllC and AllD are anchored in part by exact matching, whereas 95.4% and 97.3% of the Tit-for-Tat and Win-Stay-Lose-Shift labels are attributions made where no canonical rule fits (electronic supplementary material, §S6.2). A movement in the Tit-for-Tat share is therefore a statement about which canonical rule the play most resembles, not a claim that the model is running Tit-for-Tat.

3.5 The effect is present before any strategic interaction

A rescaling could act through the course of play, by changing how a model reads its opponent’s history, or it could act on the reading of the matrix itself. The opening move separates the two, because at round 1 nothing but the matrix and the instructions has been seen.

We find the effect already present at round 1 (figure 6). In four models of five the range of opening-move cooperation across the ten scales exceeds the range over rounds 2 to 10. The factor is 3.7 in Gemini 3.1 Flash-Lite, 2.0 in Claude Haiku 4.5, 1.9 in Gemini 3.5 Flash-Lite and 1.3 in GPT-5.4 Nano. The exception is Qwen3 235B-A22B, whose opening move sits close to 0.50 at most scales. The two windows rest on 400 and 3,600 decisions per cell, so their ranges are not variance-matched and these ratios are descriptive rather than tests.

The reading we take from this is deflationary about dynamics. Whatever the payoff scale does, most of it is already done before the game has begun. That is the timing an initial disposition read off the magnitudes of the matrix would produce. It does not rule out a further

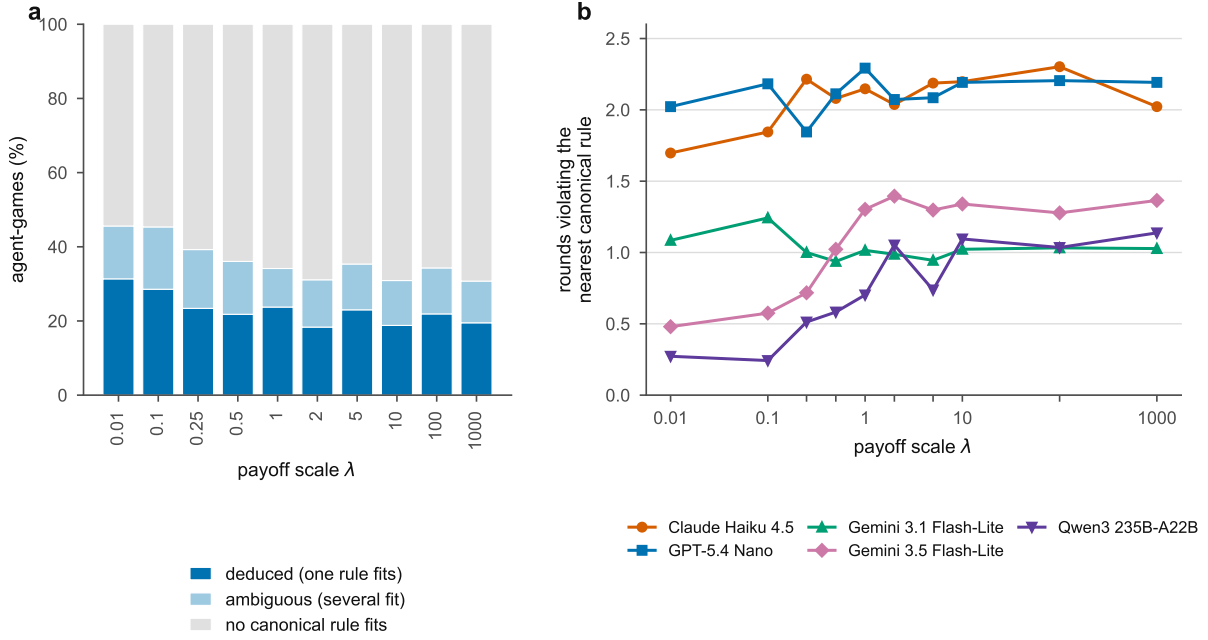


Figure 4: How closely play follows the canonical memory-one rules, against the payoff scale. (a) Provenance of the strategy read-out, pooled over the five models. Exact rule matches are less common at the largest scales than at the smallest, though the sequence is not monotone. The pooled panel is illustrative; the per-model trends, which do not all have the same sign, are in table 3. (b) Mean number of rounds, out of the ten played, violating the nearest canonical rule, computed by exact matching with no learned component.

effect on how history is weighed, and this design does not separate the two.

4 Discussion

4.1 Payoff-scale invariance is violated

The measurement reported here is unusual in having a null that is a theorem. That property is what lets a movement in cooperation be read as a departure from an invariance rather than as the difference between two conditions whose expected gap was never specified. The predicted difference between a matrix and the same matrix with larger cells is not small, or contested. It is zero.

All five models depart from it, and none is close to invariant. What makes the departures hard to set aside is not their size but their structure. They are present on the opening move, before any history exists; their size depends on the language the game is posed in; they gate the persona instruction at a boundary of magnitude; and they change what the transcripts look like as sequences.

Sampling variability alone is difficult to reconcile with this pattern, which enters at a particular round, orders itself by prompt language, turns over at a particular magnitude, and reappears in a read-out that never sees a cooperation rate. The models appear to be responding to something in the matrix other than the game it encodes, though this design does not identify what.

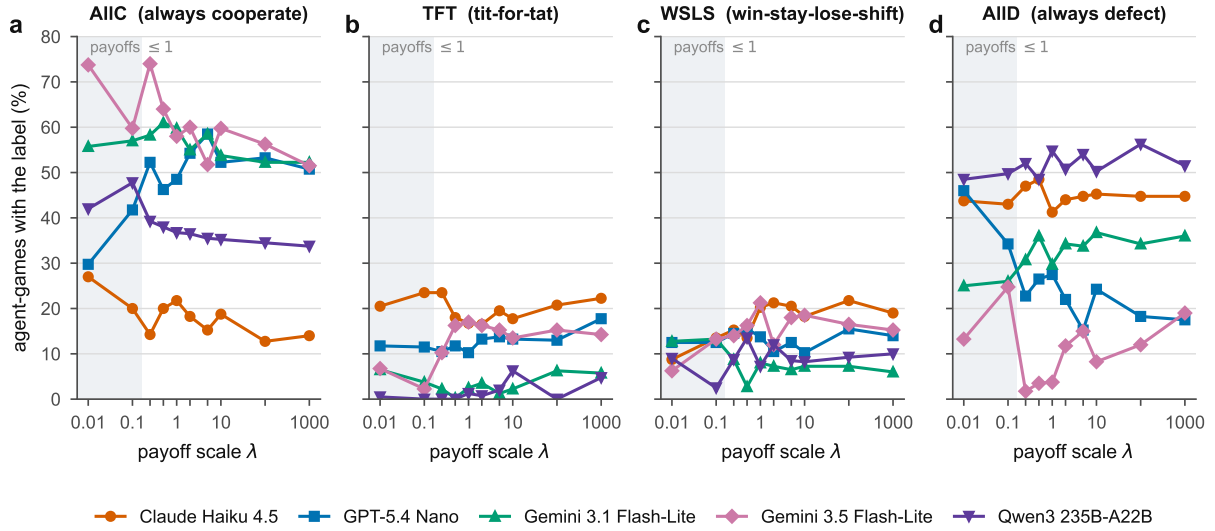


Figure 5: Share of agent-games carrying each canonical label, against the payoff scale. One panel per canonical rule: (a) AllC, (b) Tit-for-Tat, (c) Win-Stay-Lose-Shift, (d) AllD. Each point is the share of that model’s 400 agent-games at that scale carrying the label; the four panels sum to 100% and share a common vertical axis. Shaded bands mark the two scales at which every payoff printed is at most 1. The direction of movement differs between models. The AllC and AllD panels rest partly on exact rule matching, whereas the Tit-for-Tat and Win-Stay-Lose-Shift labels are 95.4% and 97.3% attributions made where no canonical rule fits, and should be read as a resemblance rather than as an identification.

4.2 Heterogeneity, and why pooled designs miss it

The pooling result deserves emphasis, because it is the mechanism by which an effect of this size could have gone unnoticed. The cause is arithmetic rather than statistical: the shapes disagree, and several pairs of them are anticorrelated. Heterogeneous effects that cancel are invisible to pooled designs by construction.

The pooled null is not a weaker version of the per-model result; it is a statement about an average model that nobody deploys. Nor is it a question of power: more dyads or more models tighten the interval around the pooled estimate without moving it, because the quantity being estimated is itself near zero. The remedy is not a larger sample but a per-model analysis.

The same logic runs one level deeper. The effect depends on the prompt language as well as on the model, so a per-model number pooled over languages is the same trick played on a smaller stage. Sensitivity to the payoff scale is a property of a model and a prompt together. The heterogeneity is not a nuisance to be averaged away. It is the finding.

4.3 Implications for behavioural benchmarking of language models

A cooperation rate published for a language model is currently understood as a property of the model. On this evidence it is a joint property of the model, the units the matrix was written in, and the language the game was posed in. The units are the awkward member of that list, because they are the one nobody is asked to record. Framings get described, personas get quoted, prompts get reproduced in appendices; the payoff scale is fixed once, silently, on the authority of a theorem saying the choice is empty.

Two laboratories running the same protocol on the same model, with matrices differing by a constant factor, could report cooperation rates differing by as much as 0.243, with no way to tell

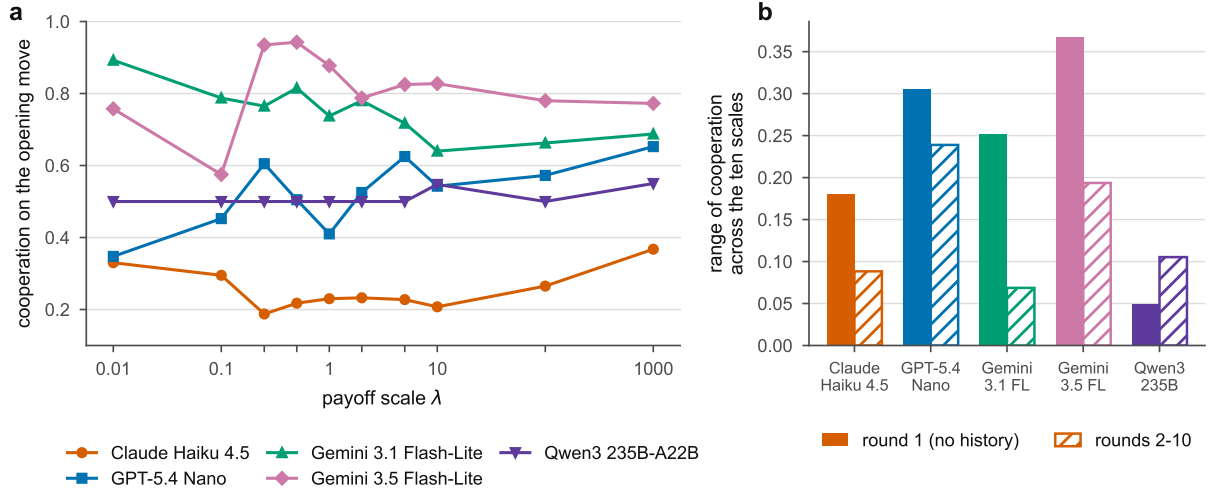


Figure 6: The scale effect is present before any history exists. (a) Cooperation on the opening move against the payoff scale; at round 1 the agent has seen only the matrix and the instructions. (b) Range of cooperation across the ten scales, on the opening move (solid) and over rounds 2 to 10 (hatched). In four models of five the opening move carries the larger range. Qwen3 235B-A22B is the exception; its opening move sits close to 0.50 at most scales.

from the numbers alone that they had not disagreed about the model.

We would suggest that a scale-sensitivity measurement accompany a published cooperation rate, in the way an error bar does, and that it be measured per model. The case is strongest where these rates leave the benchmark and become parameters: models of mixed human-machine populations take a cooperation rate as a property of the artificial agent and ask where the population settles [30, 31, 35].

4.4 Interpretation and possible mechanisms

Two observations point beyond benchmark hygiene, and both constrain what an explanation would have to account for. The first is that the boundary gating the persona instruction appears to be one of magnitude rather than of notation, and that separation is only as strong as one scale allows, because every sub-unit scale in this grid is decimal-printed too, so the two are not fully crossed. The second is that the payoff scale moves how closely play tracks any canonical memory-one rule at all, and shifts the mix of labels with it. The scale is not turning a single dial marked cooperativeness; it is changing the shape of the trajectory.

Several accounts are consistent with both. The printed magnitudes may be read as stakes, which is exactly what a rescaling does not change; numerals of different magnitudes are tokenized differently, so a representation that changes with the digit string could change what follows from it with no notion of stakes involved; training text may associate large quantities with caution or formality; and sub-unit payoffs may fall outside the range in which payoff-like numbers are ordinarily met.

We can order none of these. The design measures departures from an invariance; it does not manipulate tokenization, vary notation at fixed magnitude beyond the one contrast above, or move the framing that supplies the stakes. Nothing here should be read as identifying a mechanism.

There is a constructive reading as well as a cautionary one. A programme is under way to

widen the utility function beyond the economic consequences of an action, so that the language describing the action enters it directly, with sentiment analysis proposed as the measurement and the replicator dynamics named as the destination [29]. Our result sits one level below that programme rather than beside it. The evidence for language-based preferences comes from manipulations that could legitimately have mattered, since a label carries connotation, whereas a positive rescaling carries none and is inert by theorem, and the behaviour still moves. A utility that mixes a payoff with a score computed from language is no longer scale-free in the payoff argument unless the two terms are rescaled together, so before such an object is carried into an evolutionary dynamic its invariance class has to be stated. What these data say is that at least one class of agents does not behave as though the payoff argument were scale-free, and that anything built to describe them will have to account for it.

4.5 Limitations and future work

This design measures departures from an invariance across a broad but finite grid, and does not identify the mechanism inside the model that produces them. Five models cannot support claims about model families or about parameter count, and what generalizes here is that the shapes differ rather than what any one of them is. The corpus is one game, one prompt template family and one horizon, so whether the same curves appear in other dilemmas or in longer play is untested.

One feature of the stimuli bounds the language comparison in particular. Two of the five templates state the payoffs as penalties while phrasing the goal as maximizing a reward, and we reproduced rather than repaired them so that the corpus matches the published protocol. Because that wording is fixed across the ten scales it cannot generate a scale effect, and the within-language sensitivities stand. It does mean that a difference in level between two languages is not attributable to the language alone.

Our reference for calling a departure large is the sampling variability of these data, through a bootstrap over dyads and a permutation null, rather than an independent repeat of the whole protocol. A measured test-retest floor, from sweeps re-run under identical conditions, is the obvious next measurement.

The strategy read-out carries a further caveat, which is why its provenance is reported wherever a label appears (§2). A strategy is a decision rule and not a decision, so it is never observed directly, and even where the resident rule is fixed by construction, separating the memory-one rules from observed play has been shown to need a long run of observations [40]. Our trajectories are ten rounds long. The claim that play becomes less rule-governed rests on exact matching and no learned component. The classifier was validated to 20% execution noise, and the two models with the largest compositional movements play at close to that deviation from any rule, so the read-out is least reliable exactly where it moves most. The compositional claim uses labels the classifier supplies wherever the rules leave the answer open, is unevenly anchored across the four rules, and is the weaker of the two. Any reading of it as evidence that these models implement Tit-for-Tat would be unsupported by our data.

5 Conclusion

A positive rescaling of a payoff matrix is the rare manipulation whose predicted effect is not merely small but exactly zero. Five frontier language models were played across ten such rescalings, and every one of them moved. The movement is not a uniform bias a calibration could absorb: its shape differs by model and its size by prompt language, so the effect very nearly vanishes from any analysis that averages models, which is how a departure of this size can sit inside a literature that has not reported it.

What follows is a question about design rather than a verdict on any model. A cooperation rate measured on a language model is reported as though the matrix behind it were an arbitrary and therefore harmless choice. It is arbitrary, and it is not harmless. The same argument extends to every invariance the theory supplies, from shifts of the payoffs to relabelling of the actions, and each is an equally cheap test of whether an agent is playing the game it was shown.

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Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. The analysis pipeline that produces every figure, table and quoted number in this article, together with the tidy tables it writes and the scripts that draw the figures, is archived at [Zenodo DOI to be inserted at acceptance]. The transcript corpus of language-model games, held as per-game records of the decisions, payoffs and conditions of every dyad, and the synthetic corpus used to train the strategy classifier, which contains no language-model output, are archived at the same location under [licence to be named]. Raw per-turn request and reply text was not retained. Within the archive the three components are `Analysis/scaling/`, `Dataset/data_fairgame_frontier_llm/` and `Dataset/noise_dataset/`. Supplementary material is available online.

Declaration of AI use. Language models are the object of study in this work: the transcripts analysed here were generated by the five models named in §2, through a hosted model-serving platform. Separately from that, we used a large-language-model coding assistant to help implement and refactor the analysis pipeline and to assist with language editing of the manuscript. All analytical choices, statistical specifications and scientific interpretations are the authors'. Every number reported in this article is produced by the archived pipeline and checked against the data by an automated verification script (`Analysis/scaling/s08_verify_paper.py`), and no text, figure or result was generated by an AI system without author verification.

Authors' contributions. All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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References

- [1] Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. doi: 10.1145/3586183.3606763.
- [2] Allan Dafoe, Yoram Bachrach, Gillian Hadfield, Eric Horvitz, Kate Larson, and Thore Graepel. Cooperative AI: machines must learn to find common ground. *Nature*, 593:33–36, 2021. doi: 10.1038/d41586-021-01170-0.
- [3] Marcel Binz and Eric Schulz. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6):e2218523120, 2023. doi: 10.1073/pnas.2218523120.
- [4] John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? NBER Working Paper 31122, National Bureau of Economic Research, 2023.
- [5] Gati V. Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 337–371, 2023.
- [6] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O. Jackson. A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences*, 121(9):e2313925121, 2024. doi: 10.1073/pnas.2313925121.
- [7] Robert Axelrod and William D. Hamilton. The evolution of cooperation. *Science*, 211(4489):1390–1396, 1981. doi: 10.1126/science.7466396.
- [8] Robert Axelrod. *The Evolution of Cooperation*. Basic Books, New York, 1984.
- [9] Martin A. Nowak and Karl Sigmund. Tit for tat in heterogeneous populations. *Nature*, 355: 250–253, 1992. doi: 10.1038/355250a0.
- [10] Martin A. Nowak and Karl Sigmund. A strategy of win-stay, lose-shift that outperforms tit-for-tat in the Prisoner’s Dilemma game. *Nature*, 364:56–58, 1993. doi: 10.1038/364056a0.
- [11] Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006. doi: 10.1126/science.1133755.
- [12] William H. Press and Freeman J. Dyson. Iterated Prisoner’s Dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109(26):10409–10413, 2012. doi: 10.1073/pnas.1206569109.
- [13] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 9: 1380–1390, 2025. doi: 10.1038/s41562-025-02172-y.

- [14] Nicolò Fontana, Francesco Pierri, and Luca Maria Aiello. Nicer than humans: How do large language models behave in the prisoner’s dilemma? In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, volume 19, pages 522–535, 2025. doi: 10.1609/icwsml.v19i1.35829.
- [15] Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14:18490, 2024. doi: 10.1038/s41598-024-69032-z.
- [16] Steve Phelps and Yvan I. Russell. The machine psychology of cooperation: can GPT models operationalize prompts for altruism, cooperation, competitiveness, and selfishness in economic games? *Journal of Physics: Complexity*, 6(1):015018, 2025. doi: 10.1088/2632-072X/ada711.
- [17] Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 8086–8098, 2022. doi: 10.18653/v1/2022.acl-long.556.
- [18] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models’ sensitivity to spurious features in prompt design, or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.
- [19] Ameet Deshpande, Vishvak Murahari, Tanmay Rajpurohit, Ashwin Kalyan, and Karthik Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1236–1270, 2023. doi: 10.18653/v1/2023.findings-emnlp.88.
- [20] Leonard Salewski, Stephan Alaniz, Isabel Rio-Torto, Eric Schulz, and Zeynep Akata. In-context impersonation reveals large language models’ strengths and biases. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [21] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023. doi: 10.1017/pan.2023.2.
- [22] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, Princeton, NJ, 1944.
- [23] R. Duncan Luce and Howard Raiffa. *Games and Decisions: Introduction and Critical Survey*. Wiley, New York, 1957.
- [24] György Szabó and Gábor Fáth. Evolutionary games on graphs. *Physics Reports*, 446(4–6): 97–216, 2007. doi: 10.1016/j.physrep.2007.04.004.
- [25] Matjaž Perc and Attila Szolnoki. Coevolutionary games—a mini review. *BioSystems*, 99(2): 109–125, 2010. doi: 10.1016/j.biosystems.2009.10.003.
- [26] Nikoleta E. Glynatsi, Alex McAvoy, and Christian Hilbe. Evolution of reciprocity with limited payoff memory. *Proceedings of the Royal Society B: Biological Sciences*, 291(2025), 2024. doi: 10.1098/rspb.2023.2493.

- [27] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979. doi: 10.2307/1914185.
- [28] Amos Tversky and Daniel Kahneman. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458, 1981. doi: 10.1126/science.7455683.
- [29] Valerio Capraro, Roberto Di Paolo, Matjaž Perc, and Veronica Pizziol. Language-based game theory in the age of artificial intelligence. *Journal of the Royal Society Interface*, 21(212):20230720, 2024. doi: 10.1098/rsif.2023.0720.
- [30] Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W. Crandall, Nicholas A. Christakis, Iain D. Couzin, Matthew O. Jackson, Nicholas R. Jennings, Ece Kamar, Isabel M. Kloumann, Hugo Larochelle, David Lazer, Richard McElreath, Alan Mislove, David C. Parkes, Alex Pentland, Margaret E. Roberts, Azim Shariff, Joshua B. Tenenbaum, and Michael Wellman. Machine behaviour. *Nature*, 568:477–486, 2019. doi: 10.1038/s41586-019-1138-y.
- [31] Fernando P. Santos, Francisco C. Santos, and Jorge M. Pacheco. Social norm complexity and past reputations in the evolution of cooperation. *Nature*, 555:242–245, 2018. doi: 10.1038/nature25763.
- [32] The Anh Han, Luís Moniz Pereira, and Francisco C. Santos. Intention recognition promotes the emergence of cooperation. *Adaptive Behavior*, 19(4):264–279, 2011. doi: 10.1177/1059712311410896.
- [33] The Anh Han, Luís Moniz Pereira, Francisco C. Santos, and Tom Lenaerts. Good agreements make good friends. *Scientific Reports*, 3:2695, 2013. doi: 10.1038/srep02695.
- [34] Matjaž Perc, Jillian J. Jordan, David G. Rand, Zhen Wang, Stefano Boccaletti, and Attila Szolnoki. Statistical physics of human cooperation. *Physics Reports*, 687:1–51, 2017. doi: 10.1016/j.physrep.2017.05.004.
- [35] Ana Paiva, Fernando P. Santos, and Francisco C. Santos. Engineering pro-sociality with autonomous agents. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32, 2018. doi: 10.1609/aaai.v32i1.12215.
- [36] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. FAIRGAME: A framework for AI agents bias recognition using game theory. In *ECAI 2025 - Proceedings of the 28th European Conference on Artificial Intelligence*, pages 4097–4104. IOS Press, 2025. doi: 10.3233/FAIA251300.
- [37] Author One and Author Two. Payoff scaling shapes cooperation in LLM agents across languages, 2026. Preprint by the present authors.
- [38] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York, 1993.
- [39] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

- [40] Minjae Kim, Jung-Kyoo Choi, and Seung Ki Baek. Win-stay-lose-shift as a self-confirming equilibrium in the iterated Prisoner’s Dilemma. *Proceedings of the Royal Society B: Biological Sciences*, 288(1953):20211021, 2021. doi: 10.1098/rspb.2021.1021.